BSc (Hons) Software Engineering

University of Bedfordshire

Contextual Report

Of

Havelock Smart Resident Portal: An AI-driven Apartment Management System

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Chapter 01: Introduction

1.1 Project Background

In current days, most resident apartments in Sri Lanka are facing difficulties in communication with management of the apartment. The modern apartment communities rely on effective communication between residents, security, reception and management. Yet, a lot of residential apartments continue to handle guests, complaints, parcels, and billing using manual or partially digitalized methods.

The proposed project, named "Havelock Smart Resident Portal", is to implement a user-friendly web application for residents living in Havelock City apartment to make it easy to handle things in havelock city. Currently the havelock city mostly goes with paperwork for visitor management and billing payments. This proposed project mainly focused to reduce paperwork and to reduce delaying time for visitor pass.

This proposed project integrates AI technologies for complain prioritizing system, offer 24-hour digital passing system, online billing payments, booking inter-vehicles and parcel tracking system into a one unique system. The contextual report of the proposed project will detail the project's background, aims and objectives, description of the artefact, methodology, limitations and future scope.

1.2 Project Aims and Objectives

The aim of the proposed project is to develop an AI-driven Apartment Management System with applied security gate control that increases communication, automation and safety within the Havelock City apartments.

The objectives are,

- To design and develop a smart web-based portal integrating resident, reception, security and admin functionalities.
- To develop an AI-based complaint prioritization system.
- To create digital visitor management with 24-hour visitor pass.
- To apply online billing and tracking for transparency.

1.3 Description of the Artefact

1.3.1 Artefact

The artefact of the proposed project is to develop a web application for Resident Management System of Havelock City apartments with an AI prioritizing complaints, billing payment, parcel tracking and notifications.

The features of the Artefact of the proposed project are as follows.

- Resident, Admin, Reception and Security Dashboard.
- An AI complaint classification. (Urgent, Medium, Low)
- 24-hour visitor pass generation.
- Online billing and payment tracking.
- Inter-vehicle booking system.
- Parcel tracking and delivery notifications.
- Secure authentication and role-based access.

1.3.2 Project Design

Planning and Research (Month 01):

- Conduct a Survey to understand more about the demands and needs of users.
- Look over present techniques to identify gaps and possible spots for development.
- Describe what is exactly required, such as complaint prioritizing system and visitor management.

Design and Implementation (Month 02-04):

- Work on creating a user-friendly experience for complete comfort of use during collaboration.
- Use Node.js for starting backend development, focusing on features like complaint management system, visitor management system, online billing system and inter-vehicle booking system in MongoDB database to hold user data.
- Make sure the frontend is user-friendly and responsive by implementing it with React.js.
- Connect the frontend with the backend.

Testing and Feedback (Month 05):

- Functional testing checks whether every component such as complaints, billing, visitor pass, parcel tracking and dashboards worked as expected and free of bugs.
- Integration testing provides smooth communication between the frontend, backend and database with no API errors.

- User feedback will be collected from residents to analyze usability and identify areas for development.
- To improve system reliability and user experience, UI changes, bug fixes, and performance improvements are implemented based on user feedback.

Deployment and Finalization (Month 06):

- The database is linked to MongoDB Atlas to provide safe, flexible and dependable data storage.
- To keep sensible information safe, such JWT keys and database credentials, environment variables and security settings are setup.
- Final end-to-end testing is carried out on the live environment to make sure all features function properly after deployment.

1.4 Structure of The Report

The contextual report has four chapters.

1. **Introduction**
   This chapter explains the aims and objectives of the project, artefact of the project and project background.

2. **Literature Review and Market Research**
   The second chapter will be covered the literature review and market research of this report. This chapter will be discussed about the current applications related to my application and the difference between my application. Furthermore, data gathering and data analysis will be explained.

3. **Project Plan**
   The "Project Plan" chapter indicates the overview of my project, Work Breakdown Structure and Gantt chart.

4. **Planning The Artefact**
   The last chapter mentioned the methodology use for this project, implementation, design and planning.

Chapter 02: Literature Review and Market Research

2.1 Literature Review

2.1.1 Introduction

Digital technology has changed residential living by improving communication, management and service delivery in modern apartment communities. There is still a big lack of technologies that provide integrated, AI-driven resident interactions, since many property management systems focus on basic features like announcements or payment tracking. This literature review analyzes current research and technology applicable to Havelock Smart Resident Portal with a focus on AI-based complaint management, parcel tracking, digital visitor management and secure role-based access.

The review analyzes improvements in technology and user experience in residential management systems, focusing the increasing demand for real-time updates, smart handling and smooth communication between resident, security staff and reception. It also identifies limitations in current systems that lack advanced complaint classification and automated visitor pass generation. This provides a solid basis for developing an AI-powered, fully integrated resident management system that is perfect for the requirements of Havelock City apartments by analyzing current solutions and identifying these gaps.

2.1.2 Digital Community Platforms and Resident Management

According to current research, residential communities have become more in need of implementing digital platforms that better communication and speed service delivery. Smart resident portals usually focus on announcements, maintenance requests, event updates and basic interaction between residents and management. According to research on Smart Resident Portals digital tools improve resident and management interaction and decrease manual usage. Moreover, automation continuous to be implemented into smart living applications faster in order to improve availability and minimize issue handling delays. However, many digital community platforms still heavily depend on human participation which limits their impact in large residential apartments.

2.1.3 Current Apartment Management Systems and Their Limitations

Many current apartment management systems, like ApnaComplex, MyGate and similar platforms, provide digital solutions to support with routine residential activities. These applications primary focus on features like visitor logging, monthly charging and community notifications. Although they are convenient for individual residents, they have several limitations in terms of totally automated service management, intelligent issue prioritization and smooth communication.

ApnaComplex, for instance, allows residents to approve visitors and file complaints, but it doesn’t offer AI-based complaint prioritization. Management handles problems by hand which can cause delays, particularly when dealing with urgent issues like water leaks or electrical failures. In a similar vein, the application needs residents to enter details manually, and real-time updates are not always coordinated throughout departments (reception, security, management).

Figure 1. ApnaComplex

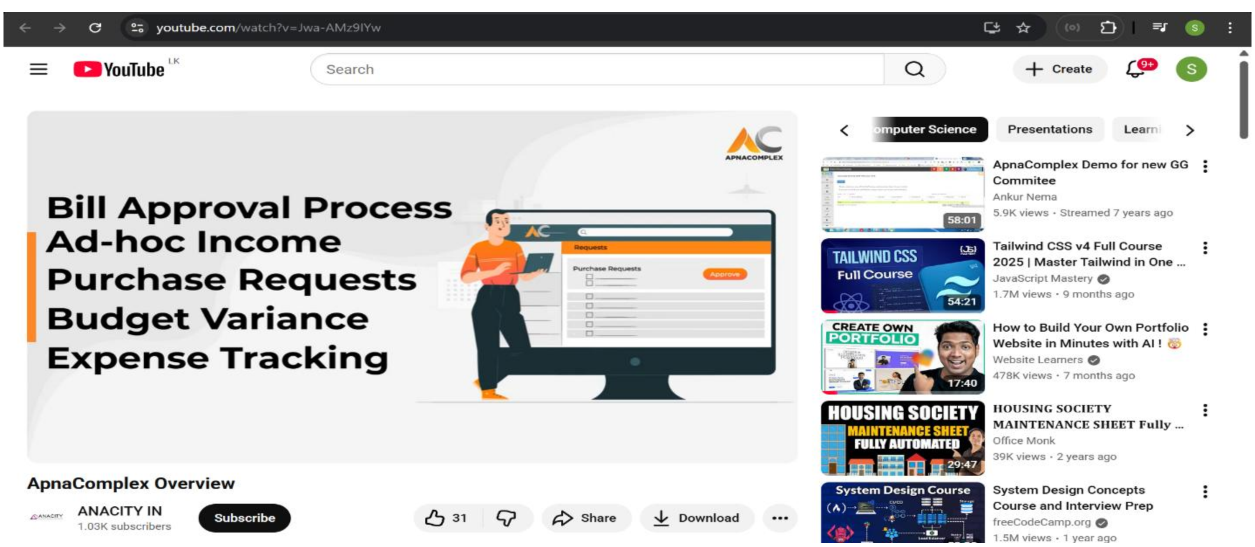

Figure 2. ApnaComplex Overview

MyGate provides advanced visitor management and gate security features, yet it also comes with limitations. It doesn’t provide a fully automated QR verification workflows or automatically generated time-limited visitor passes, but it does offer digital visitor approvals. Complaint management remains basic, without automated sorting or urgency recognition. Moreover, MyGate favors security features over unified resident management communication channels, parcel tracking and billing transparency.

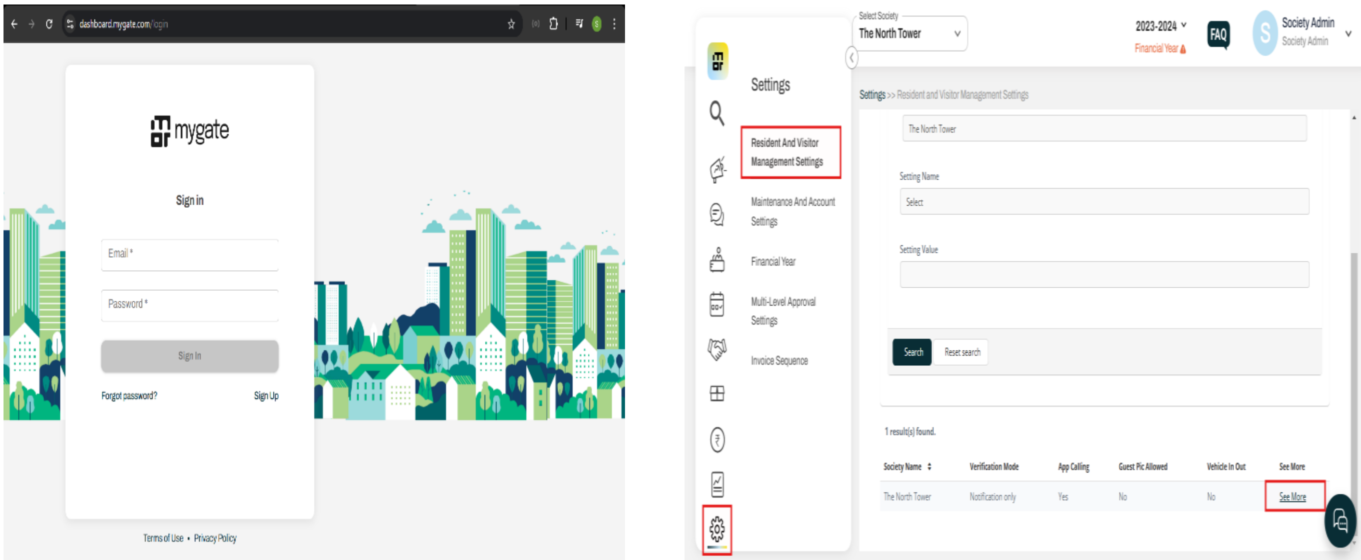

Figure 3. MyGate

Certain websites offer more sections such as facility booking or payments, however they are often disorganized and need users to navigate multiple interfaces. These platforms due to lack of centralized parcel tracking, leaving many residents are unsure of the delivery status. Moreover, none of the popular platforms offer an AI-driven decision assistance, like automated complaint classification or anticipated problem dealing, which are crucial for successful service management in big apartment buildings.

Figure 4. Other Platforms

Overall, ApnaComplex and MyGate provide fundamental online services, but they mainly designed for basic requirements of residents and don't offer a completely integrated, intelligent system. Current systems lack real-time communication between departments, AI-powered automation, interactive visitor pass generation, and single dashboards for residents, administration, reception and security. This leaves a clear need for a smart, AI-driven resident portal that delivers a smooth experience across complaints, visitors, parcels and billing, which is exactly what the Havelock Smart Resident Portal seeks to offer.

2.1.4 AI-Based Complaint Classification and Prioritization

Artificial Intelligence can categorize things, create estimates, and handle decision making processes. Because of this it has gained a lot of attention. In the field of complaint management, AI-based classification helps organizations deal with a lot of service requests by sorting out how urgent they are and classifying issues into categories more correctly than people do. Research shows that implementing rule-based systems, natural language processing and machine learning to classify complaints into levels of severity like urgent, medium and low. Applying an AI-based prioritization system breaks down on the time it takes to answer, get rids of human bias, and help to makes that the critical issues are dealt with first. Since resident portals are shown to be helpful, a majority of them still continue to depend on basic categorization which slows down the delivery of services.

2.1.5 Digital Visitor Management

Visitor management is a fundamental component of residential security. Current visitor handling methods depend on paperwork which are time-consuming and open to human error. Current research highlights the benefits of digital visitor management systems such as pre-approved visitor entry, QR-based verification and automatic pass generation. However, many residential apps currently use QR codes for quick verification features like real-time communication with security staff and auto-generated 24-hour passes are not commonly used. Furthermore, there is a poor connection between security gates and reception in existing systems which causes communication delays. These weaknesses show the need for a more automated and more effective method of visitor management.

2.1.6 Online Billing and Parcel Tracking System

Large apartment complexes are becoming progressively depend on online billing and parcel tracking. According to research online billing improves efficiency by providing people quick access to previous bills, pending payments and automatic reminders. Similarly, parcel tracking systems assist management in tracking exact delivery information and alerting residents when a parcel is delivered. However, a lot of existing technologies need a single user interface and supply key functionality as individual modules. Additionally, research shows that by decreasing doubt and decreasing delays, automated notification systems greatly improve customer happiness. Combining billing, parcel tracking and automatic notifications into one system offers residents and management staff a more better service experience.

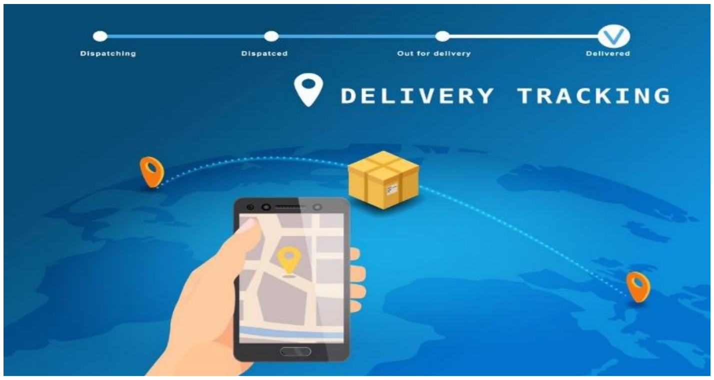

Figure 5. Delivery Tracking

2.1.7 User Experience and Role-Based Access in Smart Portals

The usefulness of role-based access control and the quality of the user experience (UX) are significant factors in the success of web-based resident portals. Research highlights that systems with simple visual layouts, simplistic dashboards and simple navigation optimize user experience and reduce training times. Role based access granted that Admin, Resident, Reception and Security receive a customized dashboard with access to just what they need. According to research implementing secure authentication techniques like JWT tokens enhances system safety and protects sensitive resident data. Considering these developments, many existing systems still don’t confirm to modern user experience standards and don’t provide customized dashboards for various user roles, leading to difficulties in daily activities.

2.1.8 Research Gap and Conclusion

The current research on smart-living technologies and digital residential management highlights a number of gaps that this project attempts to address. While many apartment management systems concentrate on fundamental features like visitor recording, maintenance billing and announcements, few focus intelligent automation or real-time communication between residents, reception and security staff. Current systems typically not support AI-driven complaint prioritization, which is crucial for maintaining that urgent issues like water leaks or power failures receive immediate attention.

Additionally, research shows that unified service portals which connect visitors, parcels, billing and complaints into a single, smooth system which have not advanced so far. Most current applications are still fragmented, residents are forced to depend on traditional communication methods like phone calls, paper notes or messaging apps. Furthermore, neither automatic visitor passes generation or real-time parcel tracking within residential communities, both of which can greatly increase convenience and transparency.

By filling these gaps through AI-driven automation, unified service management and faster communication, this project adds to the developing field of smart resident portals and gives a more efficient and user-friendly experience for modern apartment living.

Table 1. Features Between Current Systems and Proposed System

Feature	ApnaComplex	MyGate	Havelock Smart Resident Portal (Proposed Project)
AI Complaint Prioritization	No	No	Yes
24-Hour Visitor Pass	No	No	Yes
Parcel Tracking System	Yes	Yes	Yes
Online Billing	Yes	Yes	Yes
Inter-vehicle booking	No	No	Yes
Unified Single Portal	No	No	Yes
Role-based Dashboards	Yes	Yes	Yes

Major research gaps identified:

- **AI Complaint Prioritization:** Despite the fact that residents regularly report urgent issues like water leaks or power failures, current systems don't offer AI-based prioritization. Complaints are frequently addressed in the order they are received, which leads to delay in handling critical issues. Many residents difficult with getting emergency problems treated quickly due to the absence of automated urgency detection. This gap emphasizes the necessity of an intelligent complaint management system that can categorize issues according to their severity and guide management to take more effective action.

- **24-Hour Visitor Pass:** Even though some platforms provide basic visitor approvals, they don't provide real-time gate verification or automatic 24-hour visitor passes. Residents are now mainly relying on phone calls or manual entry logs, which leads to delays at the entrance gate. A digital, time-limited visitor pass system would be greatly shortened waiting times and improve security. The lack of such automated exist systems represents a significant functional gap that directly effects daily convenience for both residents and security staff.

- **Centralized Service Management:** The majority of current systems provide unconnected features, such as billing in one area, visitor approvals in another, and parcel updates via different communication channels. This dispersed strategy makes it difficult for residents to effectively handle daily activities. A unified portal that combines visits, complaints, packages and billing into a one interface is greatly desired by residents, according to survey data. An important gap in the existing apartment management systems is caused by the lack of a centralized platform, which limits efficiency and increases reliance on manual processes.

The goal of this project is to fill the major gaps present in the existing apartment management systems by developing an intelligent, fully integrated Smart Resident Portal. The system focuses on addressing major problems identified through research, such as the limitations of dispersed service management platforms, the lack of AI-based complaint prioritizing, and the lack of real-time coordination between residents, reception and security. The system develops functionality, security and communication within apartment communities with functions including automated 24-hour digital passes, AI-powered complaint urgency detection, centralized parcel tracking, and online billing.

By addressing these challenges, the proposed project delivers a more structured, responsive and user-friendly experience for both residents and management staff. This portal differs from the existing solutions like ApnaComplex and MyGate, which lack these advanced capabilities, by integrating real-time updates, integrated dashboards and intelligent automation.

In conclusion, this proposed project aims to close the gap between the growing need for smarter, more effective residential technology and conventional, traditionally-driven apartment management processes. The Havelock Smart Resident Portal offers living in apartment communities more contemporary, efficient and productive by meeting the needs for real-time communication, AI-driven prioritizing, and an all-in-one service platform.

2.2 Market Research

2.2.1 Data Gathering

Havelock Smart Resident Portal aims to give residents and management with an easy and intelligent way to manage apartment services by assisting them in improving communication, prioritizing complaints, and making informed decisions regarding visitor, billing and parcel management. A Google Forms survey consisted with 16 well-structured questions was created and shared to gather information on user preferences in order to do market research for this project. The survey was distributed online through social media, targeting people who are live in apartments.

2.2.2 Data Analysis

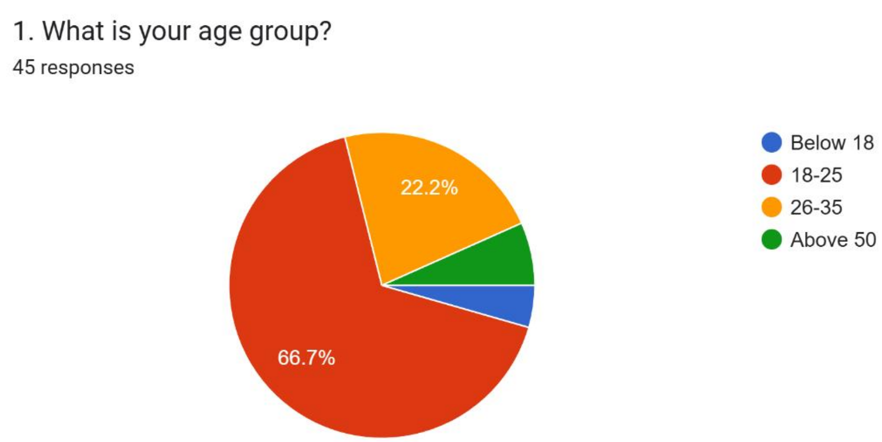

Figure 6. Questionnaire result 01

The pie chart shows with 66.7% of all participants, the 18-25 age represented the largest segment of responders across all age groups. When analyzing the results, it’s important to take this age group majority into consideration because the replies can better represent the needs and behaviors of younger residents compared to older age groups.

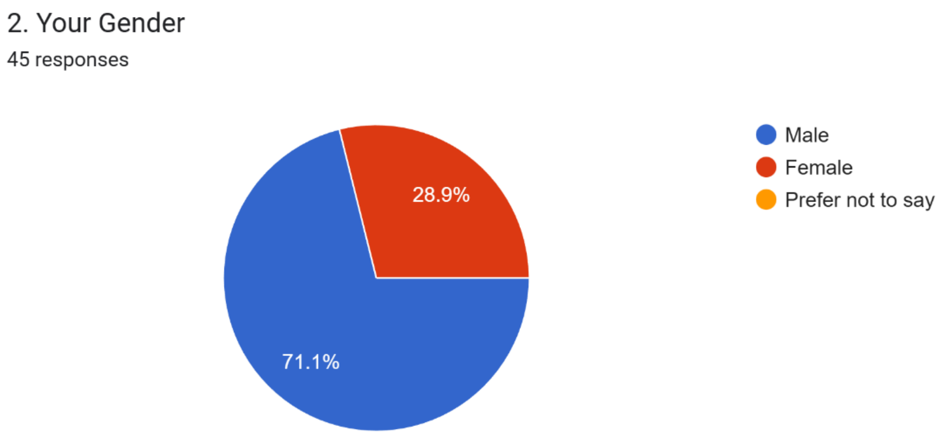

Figure 7. Questionnaire result 02

From the responses, 71.1% identified as male and 28.8% identify as female. It means that men who are live in the apartment community make up the majority of responders. The needs and want of the real user population that will engage with the smart resident portal are included in the system design, usability and feature preferences.

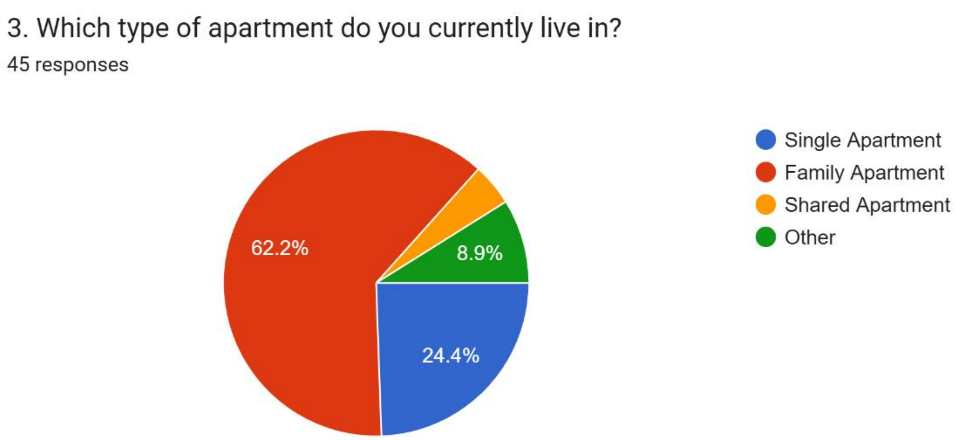

Figure 8. Questionnaire result 03

The third question indicates the participants which type of apartment they live in to better understand the living arrangements in the community. According to the results, the majority of 62.2% of respondents live in family apartment. Additionally, 4.5% living inside shared apartments and 24.4% live in single apartments. According to this collection, most users are from family-oriented residences, where efficient communication, complaint handling and visitor handling are crucial.

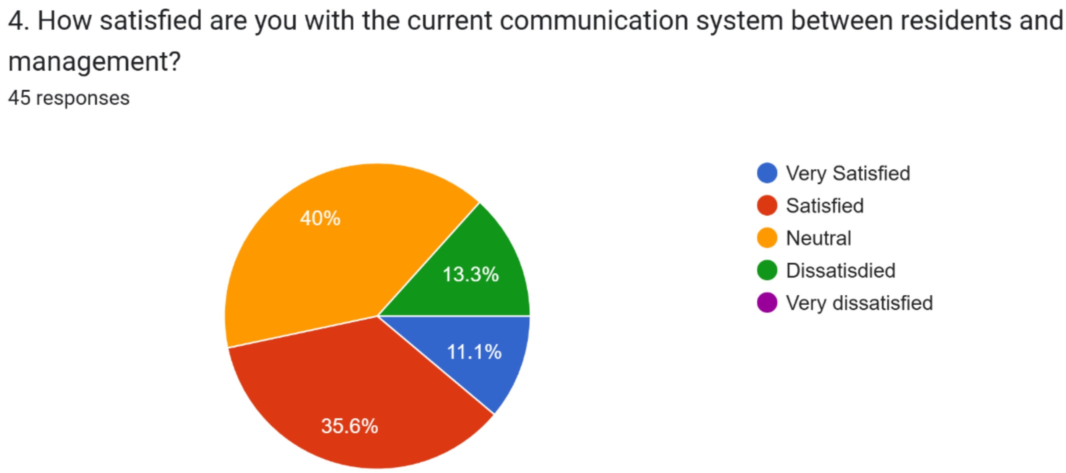

Figure 9. Questionnaire result 04

According to the respondents, 40% of participants had a neutral attitude, suggesting that many residents don't think the current communication system are either highly effective or totally insufficient. Meanwhile, 35.6% of respondents said they were satisfied and 11.1% said they were very satisfied. However, 13.3% of respondents expressed dissatisfaction pointing out significant gaps in responsiveness, clarity and accessibility. These results shows that there is a lot of space for development even though the current communication system is partly functioning.

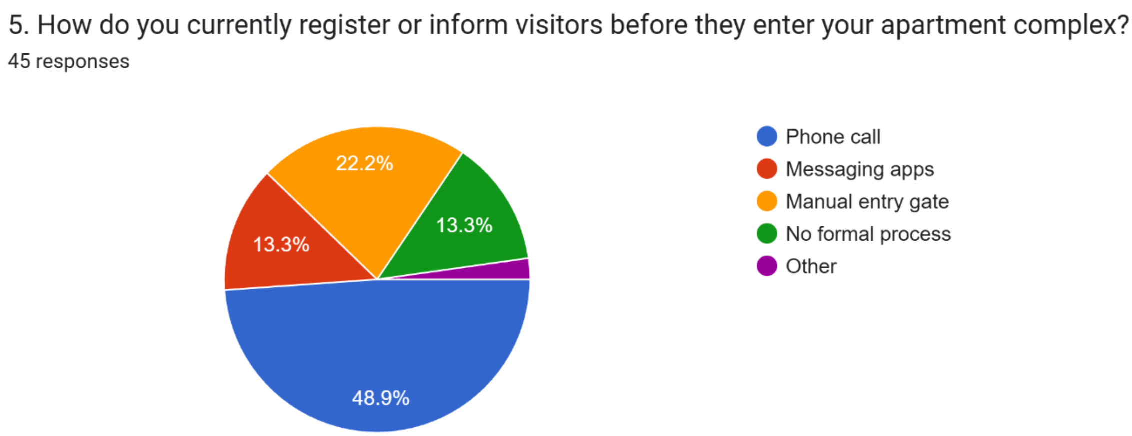

Figure 10. Questionnaire result 05

The majority of respondents, 48.9% said they communicate by phone calls which is a manual and time-consuming communication method. The reliance on conventional technologies is further shown by the fact that 22.2% use the manual entry gate procedure. Concerns regarding safety and monitoring are raised by the fact that 13.3% use messaging apps and another 13.3%

reported having no formal visitor registration process. These results clearly demonstrate the need for a more effective, digital and secure visitor management system that would allow the Havelock Smart Resident Portal's auto generated passes and real-time approvals.

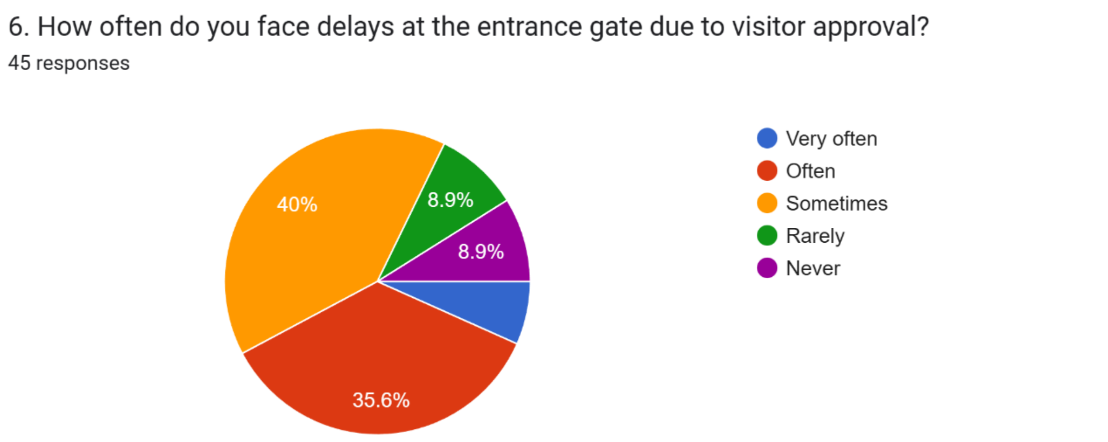

Figure 11. Questionnaire result 06

The results in the pie chart show that 40% of respondents selected "Sometimes" indicating that delays at the entrance gate due to visitor approval. And also 35.6% of respondents said they experienced delays "Often" indicates an important inefficiency in the present visitor approval process. Only a small minority reported 8.9% stated "Rarely" and another 8.9% stated "Never". However, 4.6% selected "Very often" showing that a small percentage of residents experience consistent, serious delays.

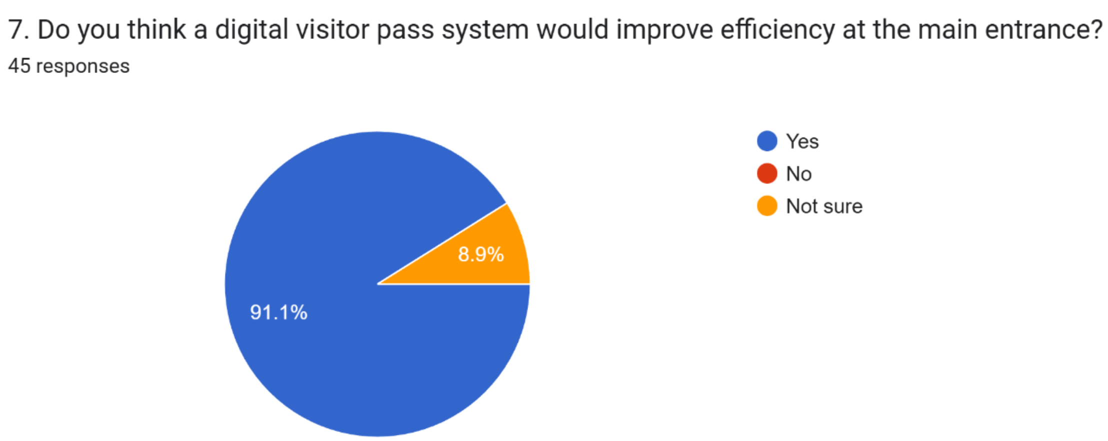

Figure 12. Questionnaire result 07

The pie chart shows the majority of residents clearly think a digital visitor pass system would improve efficiency at the main entrance. These data clearly validate the use of an automated, QR-based visitor pass within the Havelock Smart Resident Portal to avoid delays and improve security.

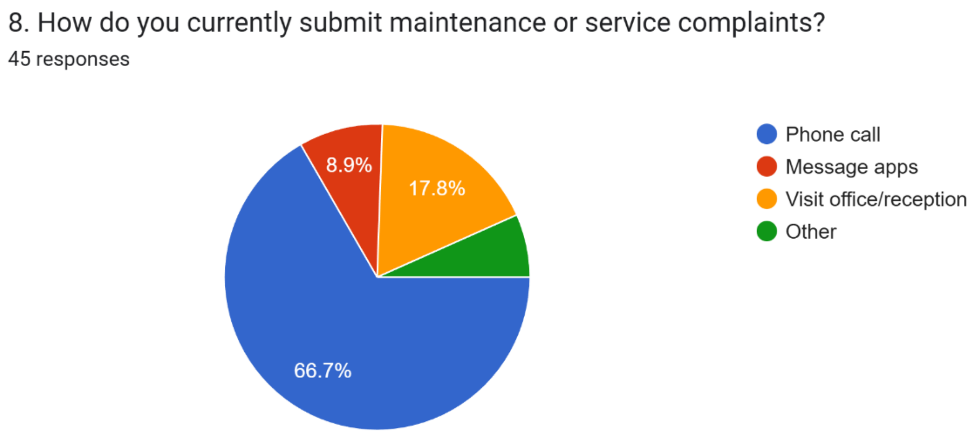

Figure 13. Questionnaire result 08

In the above pie chart, the majority 66.7% said they used phone calls, suggesting that most complaint management still depend on manual processes. Additionally, 17.8% of respondents visit the office or reception, which might result in delays and inconvenience, particularly during busy hours. These data demonstrate the lack of a centralized, efficient procedure for reporting complaints. This shows the need of the AI-driven complaint module of the Havelock Smart Resident Portal which provides digital submission, algorithmic prioritization and faster response times.

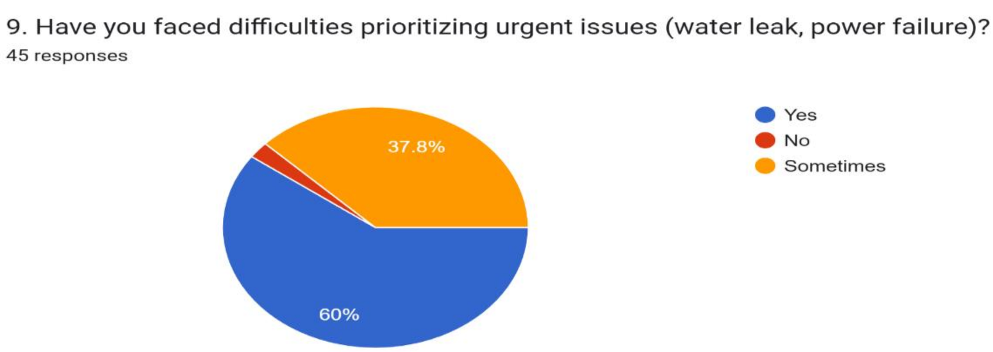

Figure 14. Questionnaire result 09

For this question, participants were questioned whether they have faced difficulties prioritizing urgent maintenance issues like water leak and power failures. According to the results, 60% of participants selected “Yes” suggesting that most residents struggle to get urgent matters resolved quickly and 37.8% of participants selected “No” suggesting that even resident who don’t frequently faced problems occasionally encounter delays or confusion in the current prioritization process.

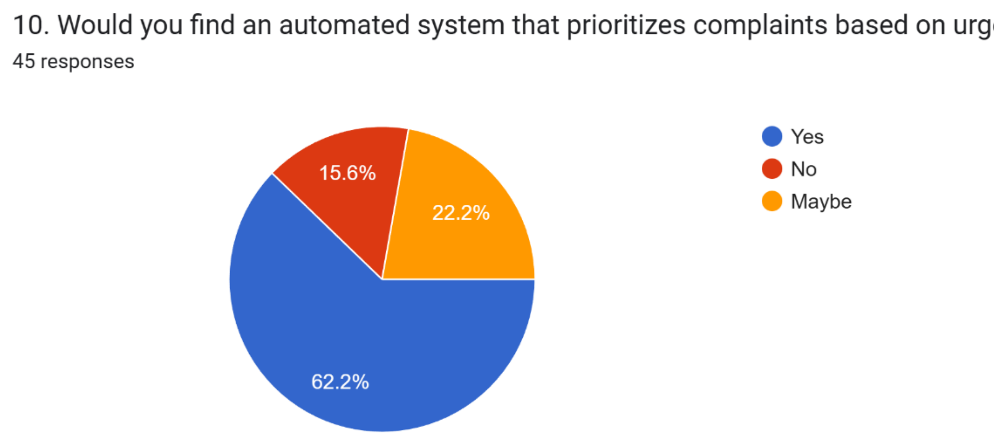

Figure 15. Questionnaire result 10

The results show that 62.2% of respondents chose “Yes” suggesting strong support for an AI-driven complaint prioritization feature, 22.2% of respondents chose “Maybe” suggesting that some respondents are open to the idea but may want to understand how the system works before using on it and 15.6% of respondents chose “No” suggesting that a small percentage may prefer traditional methods.

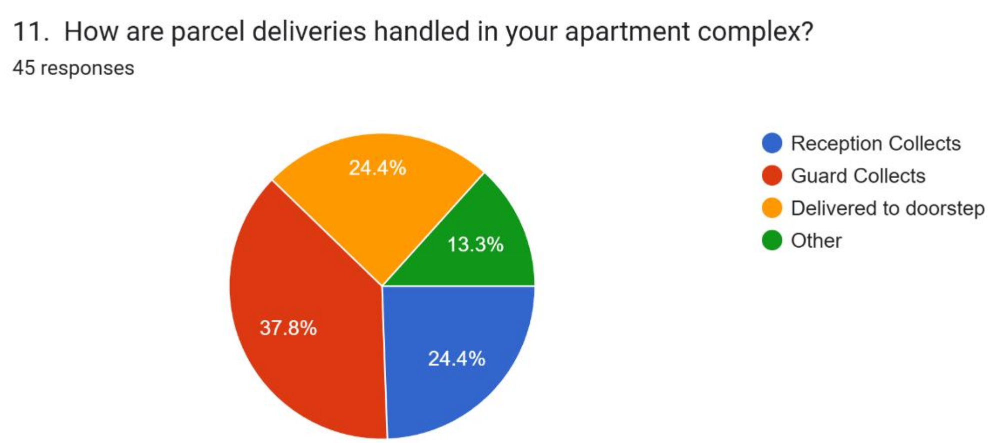

Figure 16. Questionnaire result 11

The pie chart show that 37.8% stated that parcels are collected by the security guard, making it the most unique technique. Furthermore, 24.4% reported that the reception collects parcels, while another 24.4% stated that deliveries are brought directly to their doorstep. A smaller portion, 13.3% selected “Other” suggesting variations in delivery handling depending on the situation.

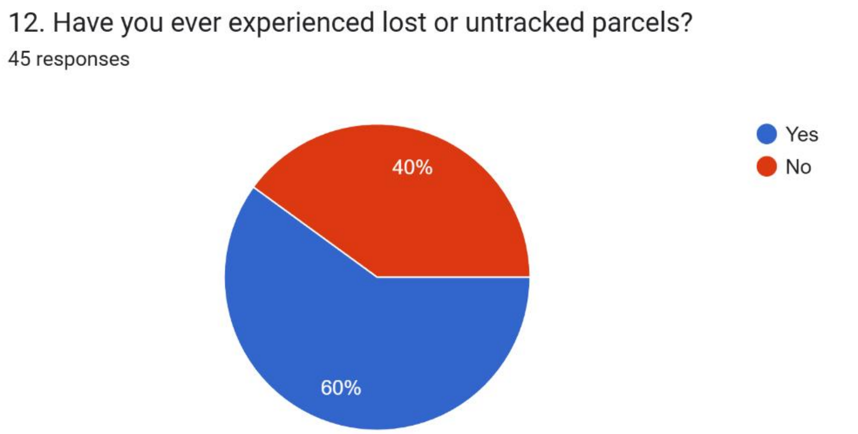

Figure 17. Questionnaire result 12

The questionnaire result show that 60% of respondents selected “Yes” suggesting that they experienced lost or untracked parcels affecting a majority of residents. However, 40% of respondents shows “No” suggesting that they never experienced lost or untracked parcels.

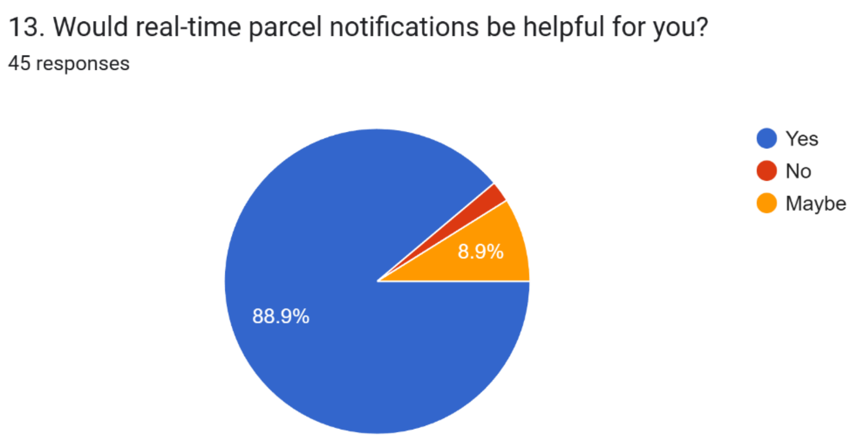

Figure 18. Questionnaire result 13

The questionnaire result show that 88.9% the majority of respondents selected “Yes” suggesting that real-time parcel notifications be helpful for them and few are thinking that real-time parcel notifications wouldn’t be helpful for them.

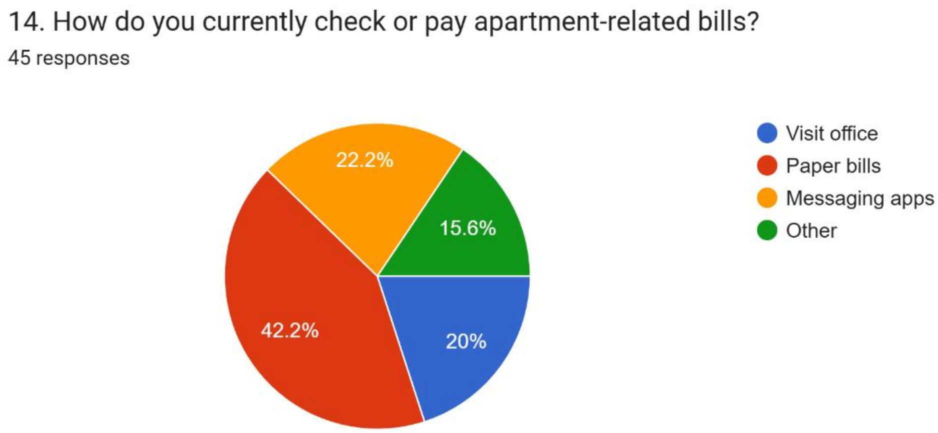

Figure 19. Questionnaire result 14

The questionnaire result show that 42.2% of residents still use paper bills, suggesting how many people still depend on traditional, non-digital ways. Furthermore, 22.2% of people use messaging apps to submit payment confirmations or receive updates, indicating informal and inconsistent contact. 20% of residents physically visit the office, which can be time-consuming and inefficient, particularly during busy hours. 15.6% of residents chose “Other” which represents variety unconventional and different approaches.

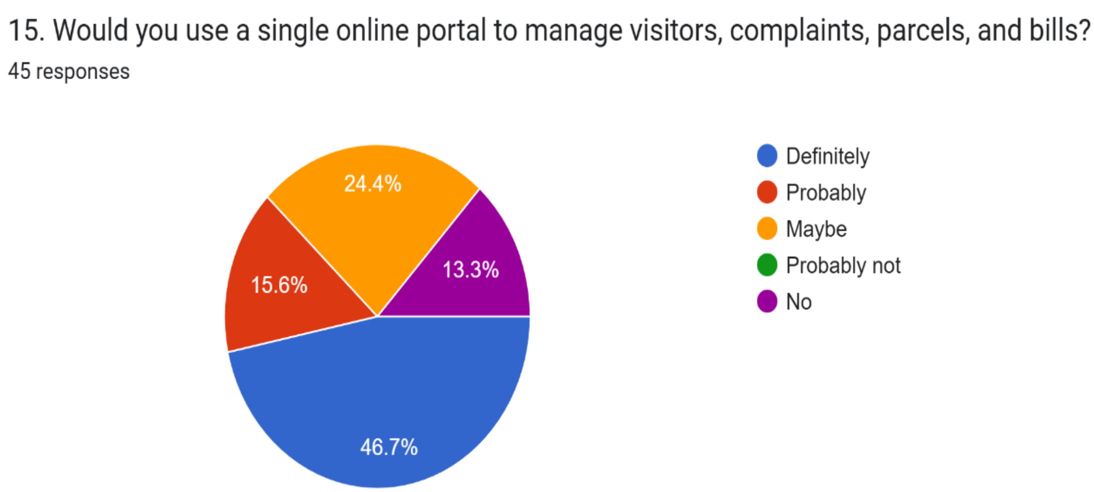

Figure 20. Questionnaire result 15

In figure 20, respondents were asked whether they would handle visitors, complaints, parcels and bills through a single online portal. According to the results, 46.7% of respondents said “Definitely” indicating a high level of interest in adopting a single online solution. Furthermore, 15.6% of respondents chose “Probably” and 24.4% chose “Maybe” suggesting that many residents are open to the idea but may want to explore the system before making a firm commitment.

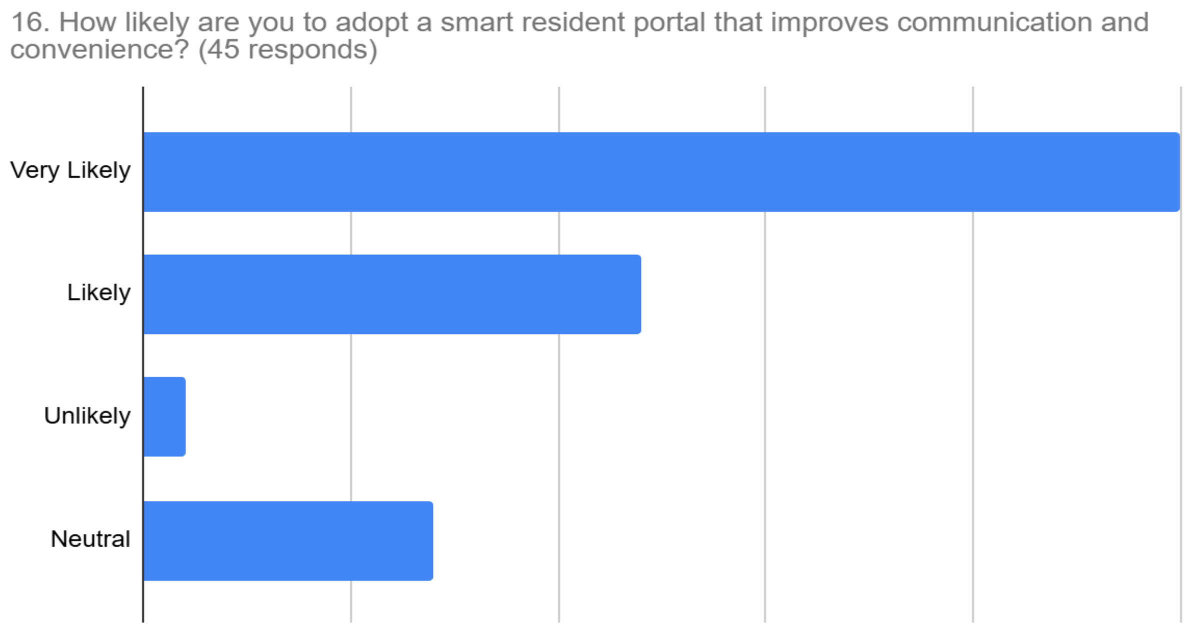

Figure 21. Questionnaire result 16

The bar chart shows a highly positive response, with 25 participants choosing “Very Likely” and 12 choosing “Likely”. Furthermore, 7 respondents selected “Neutral”, suggesting that some people could be willing to trying the method after realizing its advantages. There is only one participant selected “Unlikely”, suggesting minimal resistance to adopting new technology.

2.2.3 <u>Conclusion</u>

According to the survey responses, the majority of residents find their apartment building’s current communication and service management systems problematic. Many responses depend on traditional methods for visitor approval, parcel handling and bill payments which often result in delays, misunderstandings and inefficiencies. The data also indicates a high level of engagement in implementing digital solutions with the majority of participants thinking that features such as automated complaint prioritization, digital visitor passes, and unified service management would significantly develop their residential experience.

Residents also highlighted problems including untracked parcels, trouble prioritizing urgent complaints and inconsistent visitor management, indicating a certain need for a more organized and intelligent system. The positive response toward using a single unified portal shows the community's enthusiasm for a modern, technologically advanced solution.

By using AI-driven complaint prioritization, real-time visitor pass generation, online billing and parcel tracking, The Havelock Smart Resident Portal attempts to close these gaps and offer a smooth, practical and efficient experience for all users. This technology provides a comprehensive solution to the problems found in the survey and closely matches user experience.

Chapter 03: Project Plan

3.1 Project Overview

The Havelock Smart Resident Portal is an AI-driven web-based application designed to modernize and facilitate residential management in Havelock City apartments. The system assists residents, reception staff, security staff and management in managing day-to-day activities through intelligent automation and real-time communication. The application analyzes complaints and sets urgency levels using AI-driven complaint prioritization, ensuring that critical issues receive quick attention.

The key features of the application include AI-driven complaint prioritization, digital visitor management with auto-generated 24-hour digital passes, online billing and payment monitoring, parcel delivery notifications and role-based dashboards for all user types. Additionally, the system provides real-time updates and inter-vehicle booking, providing the entire apartment community a unified online portal.

Many residential apartments in Sri Lanka rely on manual processes like paper logbooks, phone-based complaint reporting and uncontrolled resident management communication. These traditional methods can often result in mistakes, misunderstandings and delays. The Havelock Smart Resident Portal hopes to fix these problems by providing a user-friendly, automated platform that develops transparency, security and convenience.

This system becomes a useful tool for residents and staff that want a comfortable and efficient living environment by offering AI-driven decision support, automated visitor handling and integrated service management. The goal of the project is to implement new, entirely digital solution that promotes smart living, reduces manual workload and ensures flawless daily activities in Havelock City apartments.

3.2 Work Breakdown Structure (WBS)

This Work Breakdown Structure (WBS) is help to develop the Havelock Smart Resident Portal project by breaking the work into tasks. This makes my project easier to plan and development.

The following figure shows the work breakdown structure of my project that divided into several tasks.

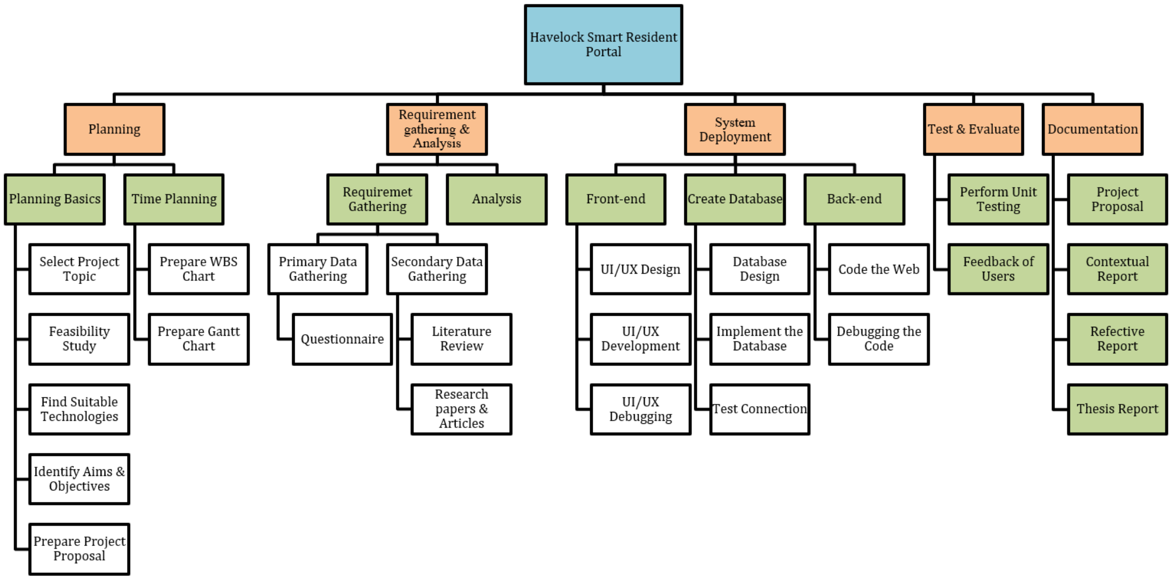

Figure 22. WBS

3.3 Gantt Chart

The figure 23 shows the Gantt chart of the proposed project.

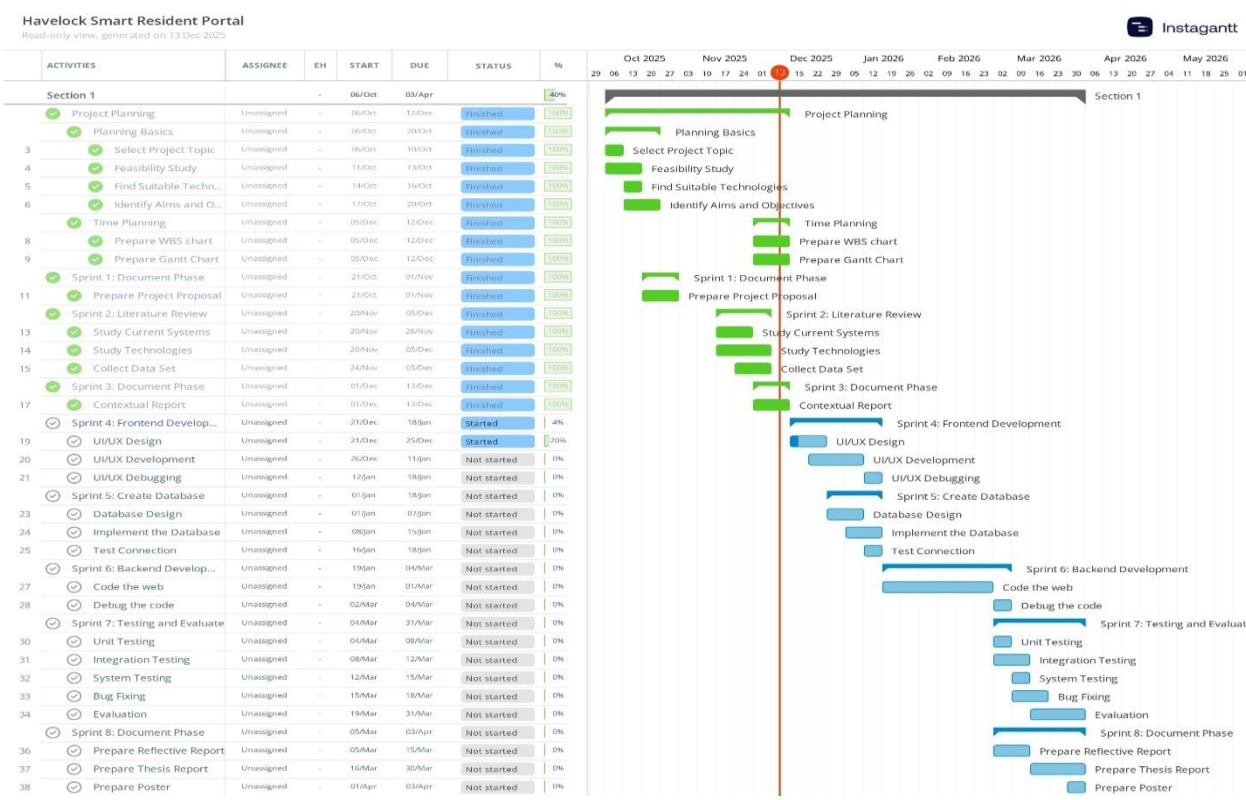

Figure 23. Gantt Chart

Chapter 04: Planning The Artefact

4.1 Methodology

Selecting a suitable development methodology is need for the successful implementation of the Havelock Smart Resident Portal. Designing and integrating these functions within the project timeline is an important task because includes developing multiple interconnected components, like AI-based complaint prioritization, 24-hour visitor pass handling, parcel tracking, online billing and role-based dashboards. The development methodology should promote ongoing updates, flexibility and continuous development based on feedback.

The Agile Development Methodology was chosen to be the ideal one for this project after reviewing various software development methodologies. The following figure 24 shows the process of the Agile Methodology.

Figure 24. The process of the Agile Methodology

4.1.1 <u>What is Agile Methodology?</u>

The Agile methodology is an updated approach to project management and software development which highlights flexibility, iteration and continuous improvement. Agile was first used in the software industry in the early 2000s and has subsequently spread into various industries as a successful method for conducting complex projects with constantly shifting needs.

The basic principle of the Agile methodology is the breaking down of big projects into smaller, more manageable units known as “sprints” which are addressed by cross-functional teamwork and continuous feedback loops. Teams can quickly react to changes, deliver incremental value, and make sure the end result satisfies clients’ changing demands thanks to this methodology.

4.1.2 Advantages of Agile Methodology in Software Development Process

The major advantages of agile methodology in software development processes, which is why it should be applied while implementing software are shown in the figure 25 and explained in detail thereafter.

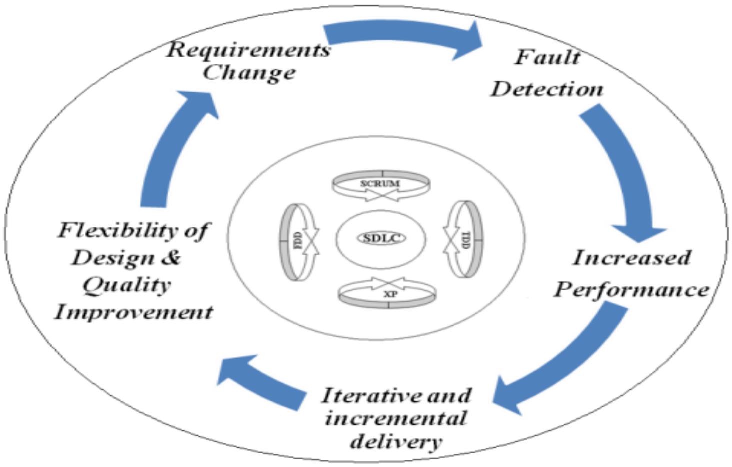

Figure 25. Agile Software Development with benefits

- **Requirements Change:** Agile develops requirement handling in the Havelock Smart Resident Portal by including residents, security, reception and management throughout development. The system can correctly depict client expectations and practical requirements via continuous feedback like modifications to visitor passes, complaint prioritization, or parcel tracking which are quickly handled.

- **Fault Detection:** Compared to a plan-driven process model, vulnerabilities are detected earlier and may be fixed before it increases in severity since testing is done at each iteration.

- **Increased Performance:** Regular standup meetings aid the Havelock Smart Resident Portal' development by allowing members to share updates, discuss issues, and develop features like visitor passes and complaint prioritization. This ongoing communication increases productivity and improves problem-solving.

- **Iterative and increment delivery:** Project delivery is split into small functional deliveries or increments to handle complexity and to get early feedback from customers and end users. These small updates used to provide on a timetable using iterations that usually last between one and four weeks each. Plans, requirements, design, code and tests initially develop and updated gradually as needed to accommodate project.

- **Flexibility of Design and Quality Improvement:** The proposed project design needs to be flexible since agile requires the application swiftly adapt to changing requirements, including visitor management update and complaint prioritization. Test-development and enhance system efficiency, increases code reuse, improves overall quality, and ensures that problems are identified and fixed effectively during every sprint.

4.2 Planning

Planning is a crucial stage in the design of the proposed project and is implemented using a work breakdown structure (WBS) and a Gantt chart. The WBS is used to breakdown the overall project into smaller, manageable tasks like planning, requirement gathering and analysis, system development, test and evaluate, and documentation. Dividing the project into these organized components helps ensure that both functional and non-functional requirements are clearly recognized and addressed.

The Gantt chart is get used to organize these tasks throughout the project timeline, defining task durations, dependencies, and goals across the six-month development period. This visible picture allows effective scheduling, progress tracking and timely identification of any delays. Through the combination of WBS and the Gantt chart, the project provides clear direction, effective time management, and organized development, guaranteeing the successful and timely delivery of the proposed project.

4.3 Requirement Gathering and Analysis

4.3.1 Primary Data: Resident Survey

Primary data is received via an online survey spread among apartment residents. The survey discusses communication methods, visitor management, complaint handling, parcel tracking and billing procedures to uncover real-world problems that residents faced. This survey spread to ensure that the Havelock Smart Resident Portal aligns with those needs of the users.

4.3.2 Secondary Data: Existing systems and Literature Review

Secondary data is collected via a review of research papers and analysis of current apartment management systems like ApnaComplex and MyGate. This helps identify existing apartment management system capabilities, limitations and research gaps.

4.3.3 Functional and Non-Functional Requirements

The following table 2 shows the functional requirements and table 3 shows the non-functional requirements.

Table 2. Functional Requirements

Requirement	Priority
User Registration	High
AI Complaint Prioritization	High
Role-Based Dashboard Access	High
24-hour Digital Visitor Management System	High
Parcel Tracking and Notification	High
Online Billing System	High
Inter-vehicle Booking System	Medium
Notification System	Medium
User Feedback System	Low
PDF Download	Low

Table 3. Non-Functional Requirements

Requirement	Priority
Performance	High
User-friendly Design	High
Scalability	Medium
Maintainability	Medium
Integrity and Safety	Low

4.4 Design

The Havelock Smart Resident Portal is designed to develop communication, automation and security within resident apartments through a unified online platform. The main objective of the application design to offer a user-friendly, reliable and scalable solution that deals with common challenges faced by residents, management and security.

The frontend of the application is designed using React.js to offer a responsive and interactive user interface accessible across desktops and mobile devices. Tailwind CSS will be used for styling to make sure a clean, modern and regular design while improving development efficiency. The backend will be implemented using Node.js and Express.js, offering a scalable and successful environment for handling REST APIs, role-based access and system operations.

This application will be developed through an AI-based logic to prioritize complaints based on urgency levels, like water leaks or power failures. This make sure crucial problems are addressed promptly. Additional automated workflows are designed for 24-hour digital visitor handling, parcel tracking, online billing, and real-time communication between management and residents. The main database for safely maintaining resident data, complaints, visitor records, billing records and parcel data is MongoDB. Firebase will also integrate to support real-time notification and application updates. JWT-based authentication will be implemented to make sure safe access and protect sensitive user records.

4.5 Implementation

The main objective of the Havelock Smart Resident Portal's implementation phase is to make the proposed project design into a functional and reliable web-based application. Using current web technologies, this phase involves developing fundamental modules like user identification, 24-hour digital visitor management, complaint handling with AI-based prioritization, parcel tracking, and billing management. Agile development techniques are used in the system's implementation, enabling incremental development, regular testing, and rapid requirement-based improvement. To make sure that the application fulfills both functional and non-functional requirements while offering a practical solution for apartment management, priority is made on secure coding techniques, role-based access control, and real-time system availability.

The Table 4 indicates the software requirements that need to implement the Havelock Smart Resident Portal website application and design it.

Table 4. Software Requirements

Software	Version	Objective
React.js	Latest	Developing the Front-end.
Node.js	Latest	Providing a runtime environment and allowing to execute the JavaScript code outside of a web browser.
MongoDB	Latest	Storing user data and develop configuration.
JavaScript	Latest	For AI-based complaint prioritization.
Firebase	Latest	User authentication and real-time data storage.
Tailwind CSS	Latest	Styling the user interface.
TypeScript	Latest	Improving code quality and maintainability.
Morgan	Latest	To log details about HTTP requests and supporting to fix bugs.

The Table 5 indicates the Hardware requirements that need to implement the Havelock Smart Resident Portal Web application.

Table 5. Hardware Requirements

Component	Specification	Objective
Development Computer	Processor: Intel Core i5	For developing, testing and debugging the frontend and backend of the web application.
Internet Connection	Router, Wi-Fi access points, broadband internet	Make sure stable connectivity between users and the system.

4.6 Testing and Evaluation

The testing and evaluation process for the Havelock Smart Resident Portal system validates that the application works correctly and satisfies user needs. Individual modules like user authentication, visitor management, complaint management, parcel tracking and billing are put through to unit testing. Integration testing shows that the Node.js backend and React.js frontend are communicating well. System testing checks performance, reliability and real-world user situations, while user acceptance testing (UAT) collects feedbacks from residents and staff to develop usability and make sure the application is prepared for deployment.

Non-functional aspects like security, usability and performance are evaluated in addition to functional testing. Authentication processes are tested to provide secure role-based access, while system responsiveness is evaluated under various load conditions. In order to make sure that users with different technical skills can interact with the system, usability testing focuses on interface clarity and simplicity of navigation.

4.7 Risk and Mitigation

While implementing the Havelock Smart Resident Portal, there might be some risks that could impact the performance of the project. It’s important to recognize these risks early and then implement practical mitigation plans to reduce their effects.

The below table shows about the risk analysis and the mitigation actions that can be taken.

Table 6. Risk analysis and the mitigation actions

Risk	Impact	Likelihood	Mitigation Action
Integration failure between frontend and backend.	High	Medium	Use postman for API testing.
Delay in project completion.	High	Medium	Applying agile methodology.
Errors in AI complaint prioritization.	Medium	Medium	Use simple rule basic logic.
Database performance issues.	Medium	Medium	Use MongoDB Atlas to identify and implement proper data.
Poor user adoption.	Low	Low	Use Tailwind CSS for interface.

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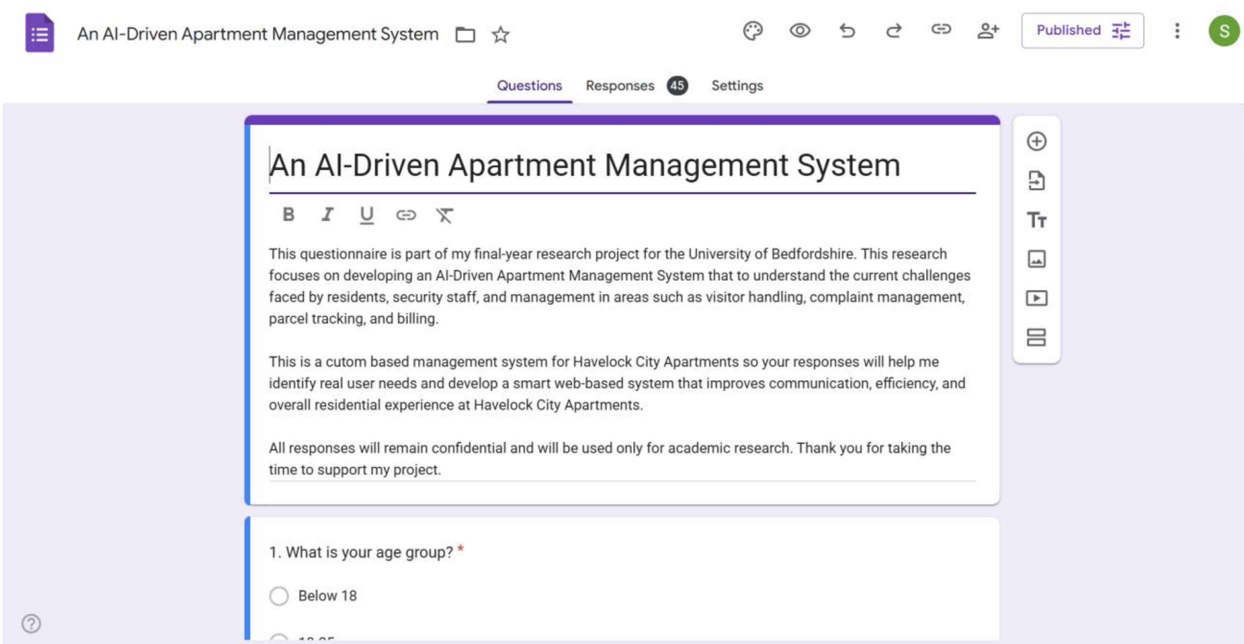
Appendix

Questionnaire



An AI-Driven Apartment Management System

Published

Questions Responses 45 Settings

An AI-Driven Apartment Management System

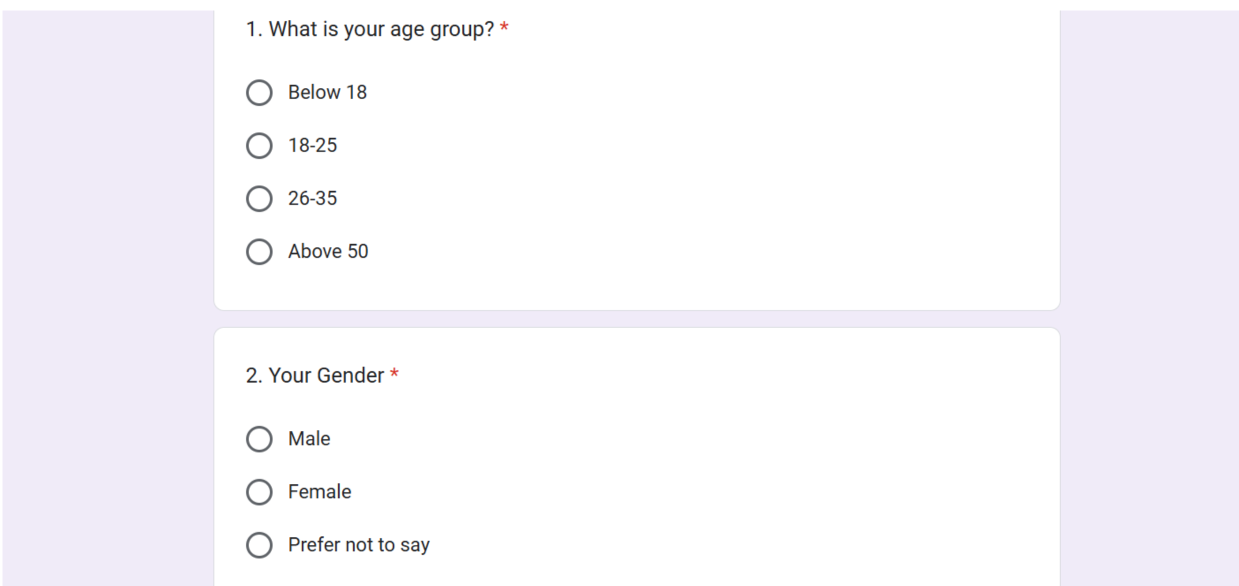
This questionnaire is part of my final-year research project for the University of Bedfordshire. This research focuses on developing an AI-Driven Apartment Management System that to understand the current challenges faced by residents, security staff, and management in areas such as visitor handling, complaint management, parcel tracking, and billing.

This is a cutom based management system for Havelock City Apartments so your responses will help me identify real user needs and develop a smart web-based system that improves communication, efficiency, and overall residential experience at Havelock City Apartments.

All responses will remain confidential and will be used only for academic research. Thank you for taking the time to support my project.

1. What is your age group? *

Below 18



1. What is your age group? *

- Below 18
- 18-25
- 26-35
- Above 50

2. Your Gender *

- Male
- Female
- Prefer not to say

4. How satisfied are you with the current communication system between residents and management? *

- ◯ Very Satisfied
- ◯ Satisfied
- ◯ Neutral
- ◯ Dissatisdied
- ◯ Very dissatisfied

5. How do you currently register or inform visitors before they enter your apartment complex? *

- ◯ Phone call
- ◯ Messaging apps
- ◯ Manual entry gate
- ◯ No formal process
- ◯ Other

6. How often do you face delays at the entrance gate due to visitor approval? *

- ◯ Very often
- ◯ Often
- ◯ Sometimes
- ◯ Rarely
- ◯ Never

7. Do you think a digital visitor pass system would improve efficiency at the main entrance? *

- ◯ Yes
- ◯ No
- ◯ Not sure

8. How do you currently submit maintenance or service complaints? *

- Phone call
- Message apps
- Visit office/reception
- Other

9. Have you faced difficulties prioritizing urgent issues (water leak, power failure)? *

- Yes
- No
- Sometimes

10. Would you find an automated system that prioritizes complaints based on urgency helpful? *

- Yes
- No
- Maybe

11. How are parcel deliveries handled in your apartment complex? *

- Reception Collects
- Guard Collects
- Delivered to doorstep
- Other

12. Have you ever experienced lost or untracked parcels? *

- Yes
- No

13. Would real-time parcel notifications be helpful for you? *

- Yes
- No
- Maybe

14. How do you currently check or pay apartment-related bills? *

- Visit office
- Paper bills
- Messaging apps
- Other

15. Would you use a single online portal to manage visitors, complaints, parcels, and bills? *

- Definitely
- Probably
- Maybe
- Probably not
- No

16. How likely are you to adopt a smart resident portal that improves communication and convenience? *

- Very Likely
- Likely
- Neutral
- Unlikely
- Very Unlikely

Submit

Clear form